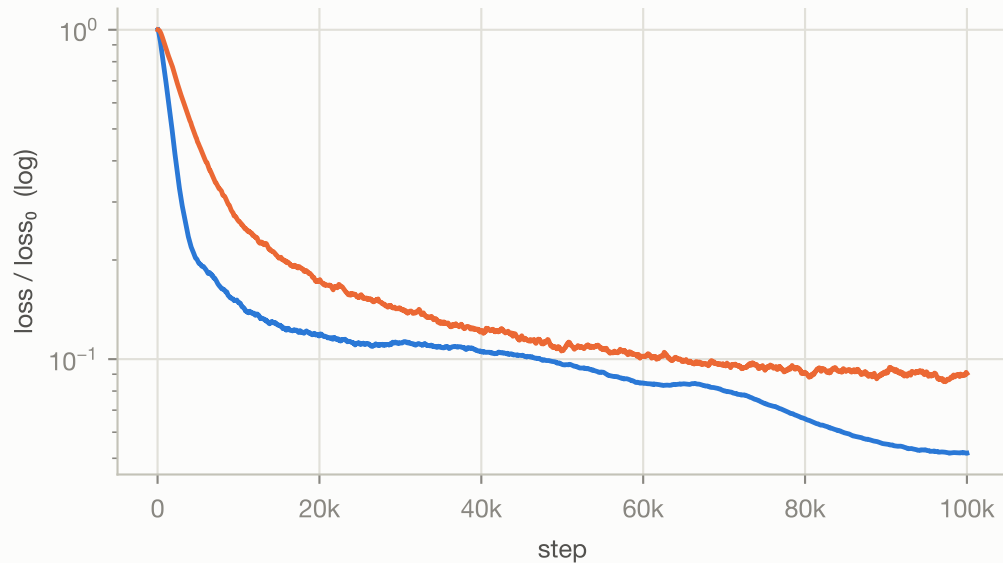


vJEPA vs vMAE — Active Matter

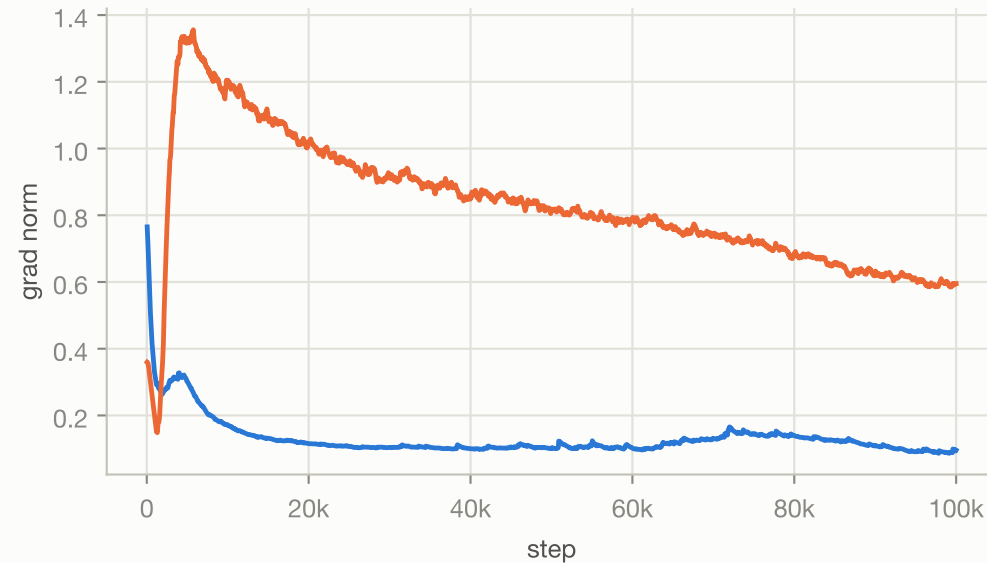
11ch · 256×256 · nematic active-matter turbulence

100,000 steps each · identical ViT-S encoder, tube masking @0.9 — wall-clock: JEPA 8.6h / MAE 9.1h — final loss (raw): JEPA 0.027 / MAE 0.117

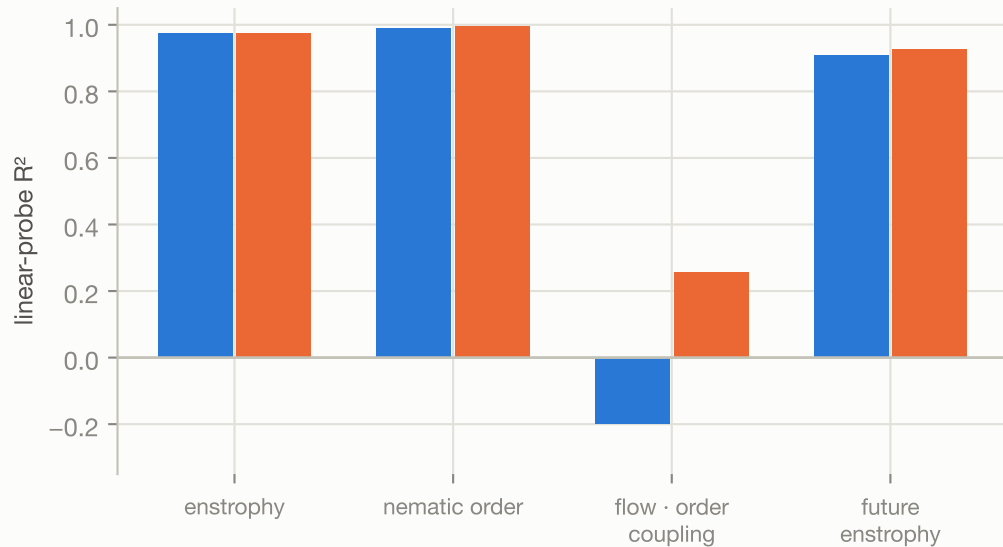
Training loss, relative to step 50



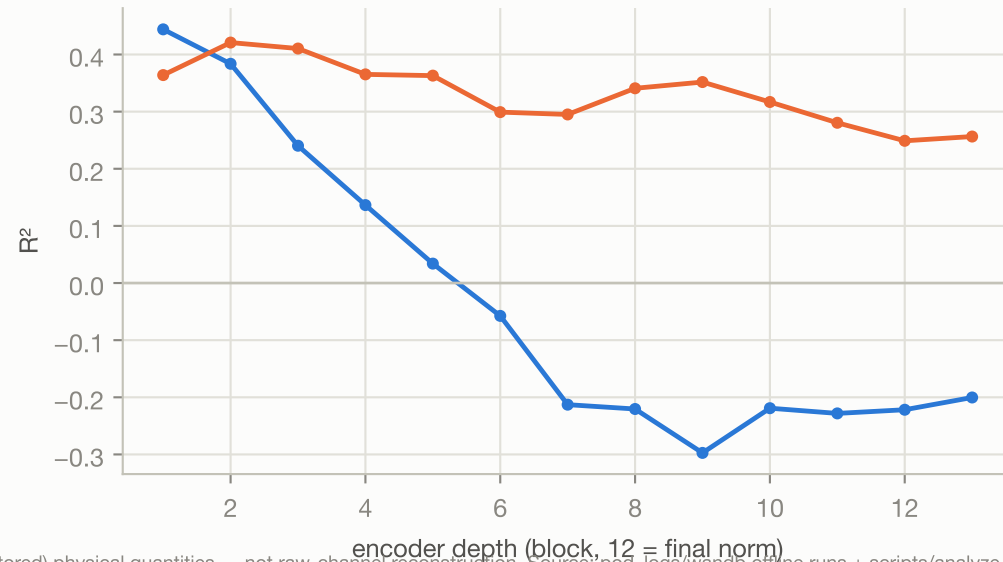
Gradient norm (training stability)



Physics decodability, final layer (ridge probe, 5-fold)



“flow · order coupling” R^2 by depth



Loss curves EMA-smoothed ($\alpha=0.06$). R^2 = 5-fold ridge probe on mean-pooled encoder features vs. derived (non-stored) physical quantities — not raw-channel reconstruction. Source: pod_logs/wandb offline runs + scripts/analyze_encoder